

Mahalanobis Distance

The Mahalanobis distance of a point $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top \in \mathbb{R}^n$ from a probability distribution:

$$d = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$$

with mean vector $\boldsymbol{\mu} \in \mathbb{R}^n$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$.

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Machine_learning	88,260	56,966
Deep_learning	36,741	15,321
Statistics	29,673	28,341
Mathematical_optimization	19,318	6,819
Data_mining	13,168	5,750
Supervised_learning	14,120	3,596
Unsupervised_learning	11,693	2,702
Computer_program	13,451	16,587
Algorithm	38,905	36,203
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Anomaly_detection	7,540	1,514

The dataset is available at <https://arxiv.org/abs/2605.16361>

$$\text{Mean vector } \boldsymbol{\mu} = \begin{bmatrix} 2.5 \times 10^4 \\ 1.6 \times 10^4 \end{bmatrix}$$

$$\text{Covariance matrix } \boldsymbol{\Sigma} = \begin{bmatrix} 4.6 \times 10^8 & 3.2 \times 10^8 \\ 3.2 \times 10^8 & 2.7 \times 10^8 \end{bmatrix}$$

Covariance Matrix $\Sigma \in \mathbb{R}^{2 \times 2}$

Eigenvalue decomposition:

$$\Sigma = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 6.98 \times 10^8$$

$$\lambda_2 = 0.28 \times 10^8$$

and eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$$

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```
1 import numpy as np
2
3 S = np.array([[4.57602874e+08,
4                3.21703384e+08], [3.21703384e
5                +08, 2.68455504e+08]])
6 eig_val, eig_vec = np.linalg.eig(S)
7
8 print('lambda_1 = ', eig_val[0])
9 print('lambda_2 = ', eig_val[1])
10 print('u_1 = ', eig_vec[:, 0])
11 print('u_2 = ', eig_vec[:, 1])
```

The squared Mahalanobis distance is an ellipse in the two-dimensional space.

$$\begin{aligned}d^2 &= (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\ &= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top \right)^{-1} (\mathbf{x} - \boldsymbol{\mu})\end{aligned}$$

The squared Mahalanobis distance is an ellipse in the two-dimensional space.

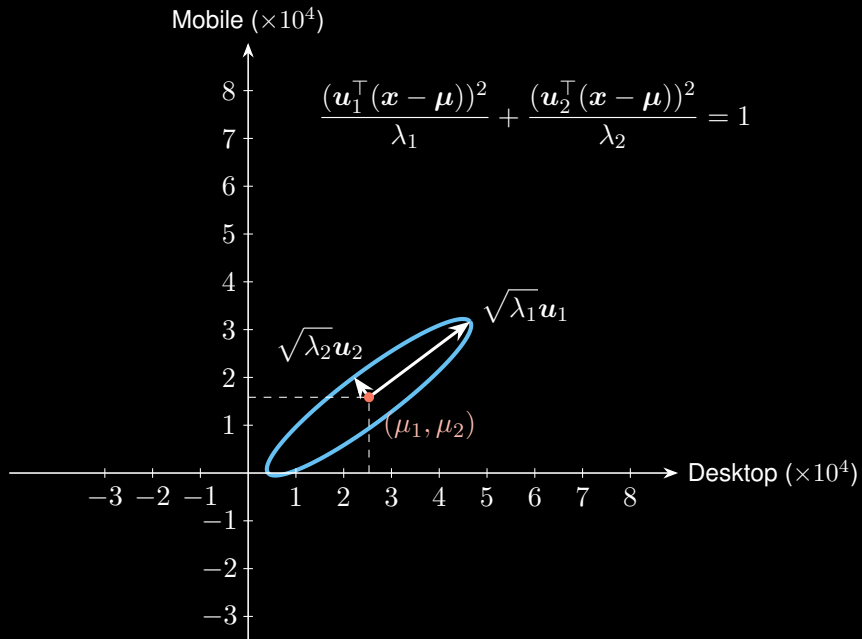
$$\begin{aligned}d^2 &= (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\&= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top \right)^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\&= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\frac{1}{\lambda_1} \mathbf{u}_1 \mathbf{u}_1^\top + \frac{1}{\lambda_2} \mathbf{u}_2 \mathbf{u}_2^\top \right) (\mathbf{x} - \boldsymbol{\mu}) = c\end{aligned}$$

The squared Mahalanobis distance is an ellipse in the two-dimensional space.

$$\begin{aligned}d^2 &= (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\&= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top \right)^{-1} (\mathbf{x} - \boldsymbol{\mu}) \\&= (\mathbf{x} - \boldsymbol{\mu})^\top \left(\frac{1}{\lambda_1} \mathbf{u}_1 \mathbf{u}_1^\top + \frac{1}{\lambda_2} \mathbf{u}_2 \mathbf{u}_2^\top \right) (\mathbf{x} - \boldsymbol{\mu}) = c\end{aligned}$$

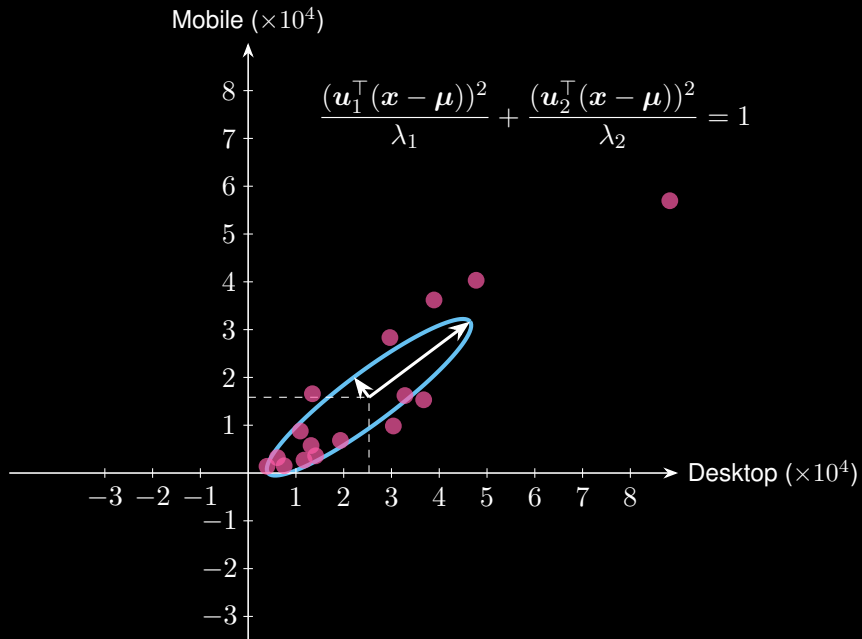
$$\Rightarrow \frac{(\mathbf{u}_1^\top (\mathbf{x} - \boldsymbol{\mu}))^2}{\lambda_1} + \frac{(\mathbf{u}_2^\top (\mathbf{x} - \boldsymbol{\mu}))^2}{\lambda_2} = d^2 \quad (\text{ellipse})$$

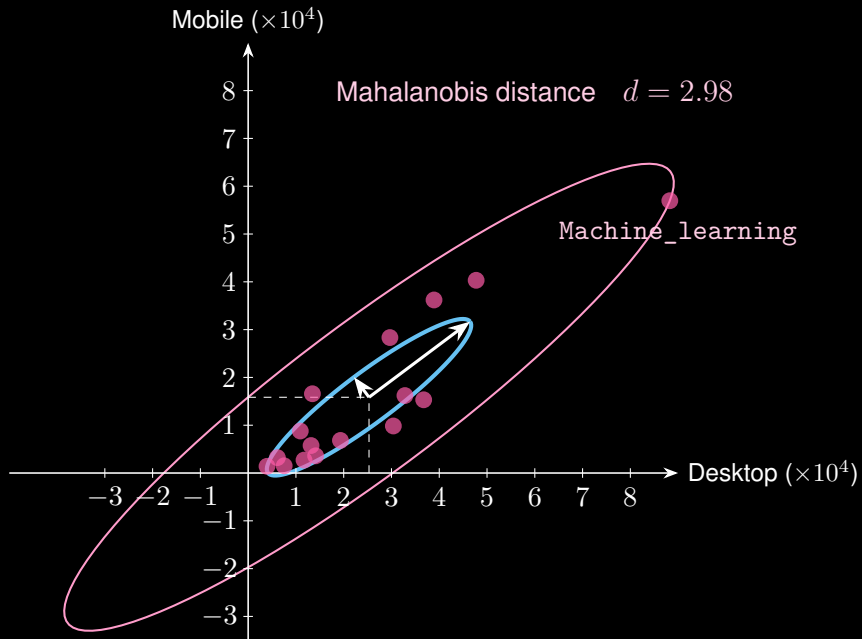
with eigenvalues λ_1, λ_2 and eigenvectors $\mathbf{u}_1, \mathbf{u}_2$.

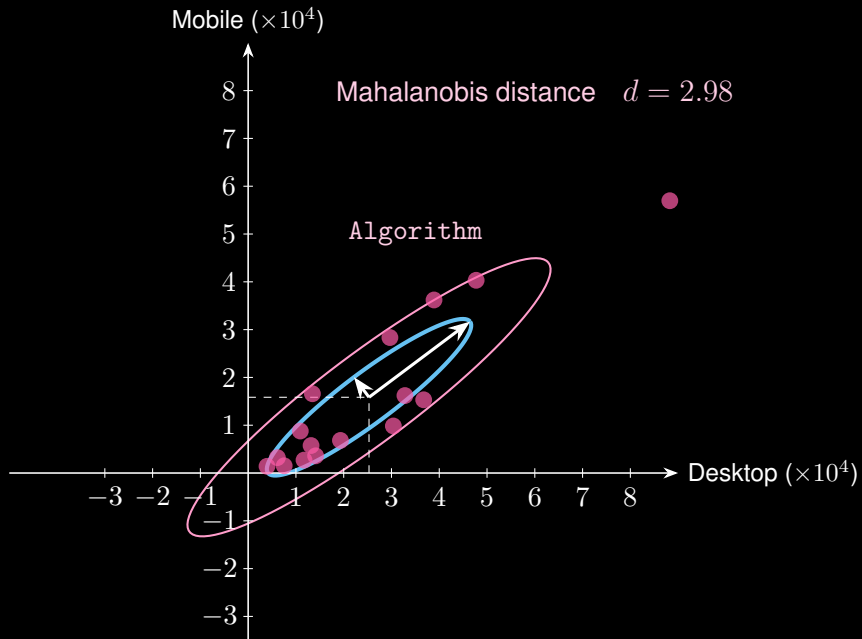


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Thanks for your attention!

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